Joint Mean-Variance Inference for Nonlinear Mixed-Effects Models with Measurement Errors and Outliers

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Overview

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- The mean and variance models for NLME
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- Measurement error model

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Nonlinear mixed-effects (NLME) models

NLME model - the mean model

$$y_{ij} = g(\boldsymbol{x}_{ij}, \boldsymbol{\beta}, \boldsymbol{u}_i) + e_{ij} \tag{1}$$

for $i = 1, \cdots, n$; $j = 1, \cdots, n_i$, where

- $g(\cdot)$: a known nonlinear function for the mean $\mathrm{E}(y_{ij})$

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- $g(\cdot)$: a known nonlinear function for the mean $\mathrm{E}(y_{ij})$

  E.g., the exponential decay NLME model for the viral load trajectories during the treatment[1]

$$y_{ij} = (\beta_1 + u_{i1}) + (\beta_2 + u_{i2})e^{-(\beta_3 + u_{i3})t_{ij}} + e_{ij}$$

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- $g(\cdot)$: a known nonlinear function for the mean $\mathrm{E}(y_{ij})$
- $\boldsymbol{\beta}$: fixed effects
- $\boldsymbol{u}_i$: random effects. $\boldsymbol{u}_i \sim \mathcal{N}(\mathbf{0}, D)$

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- $\boldsymbol{u}_i$: random effects. $\boldsymbol{u}_i \sim \mathcal{N}(\mathbf{0}, D)$
- e_{ij}: within-individual error.

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- e_{ij}: within-individual error. $e_{ij} \sim \mathcal{N}(0, \sigma_0^2)$

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- $g(\cdot)$: a known nonlinear function for the mean $\mathrm{E}(y_{ij})$
- $\boldsymbol{\beta}$: fixed effects
- $\boldsymbol{u}_i$: random effects. $\boldsymbol{u}_i \sim \mathcal{N}(\mathbf{0}, D)$
- e_{ij}: within-individual error. $\cancel{e_{ij} \sim \mathcal{N}(0, \sigma_0^2)}$

$$e_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$$

allow the within-individual variances to be <u>individual-specific</u> and <u>time-dependent</u>

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Nonlinear mixed-effects (NLME) models

Model the within-individual variances σ_{ij}^2

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Nonlinear mixed-effects (NLME) models

Model the within-individual variances σ_{ij}^2 ,
to investigate

- if the variances may be partially explained by time-varying covariates

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Nonlinear mixed-effects (NLME) models

Model the within-individual variances σ_{ij}^2 ,
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- if the variances may be partially explained by time-varying covariates
- if the variances may vary substantially across individuals

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Nonlinear mixed-effects (NLME) models

Model the within-individual variances σ_{ij}^2 ,
to investigate

- if the variances may be partially explained by time-varying covariates
- if the variances may vary substantially across individuals

NLME model - the variance model

$$\log(\sigma_{ij}^2) = \alpha_1 + \alpha_2 z_{ij} + a_i \tag{2}$$

for $i = 1, \cdots, n$; $j = 1, \cdots, n_i$, where

- z_{ij}: a time-varying covariates
- a_i: a individual-specific random effect. Assume $a_i \sim \mathcal{N}(0, \sigma_a^2)$ or $\exp(a_i) \sim k/\chi_k^2$

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The advantages of modeling the variance

- Allow for inconstant variance, better understand the nature of the within-individual variation

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The advantages of modeling the variance

- Allow for inconstant variance, better understand the nature of the within-individual variation

  E.g. Understanding the nature of the within-individual variations in viral load allows us to choose good timing for treatment interruption

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- Allow for inconstant variance, better understand the nature of the within-individual variation
- Obtain more efficient estimates (i.e., with smaller SE) of the main parameters of interest

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- Allow for inconstant variance, better understand the nature of the within-individual variation
- Obtain more efficient estimates (i.e., with smaller SE) of the main parameters of interest

E.g., model the variance in the NLME model for the viral load

$$y_{ij} = (\beta_1 + u_{i1}) + (\beta_2 + u_{i2})e^{-(\beta_3+u_{i3})t_{ij}} + e_{ij}$$

Obtain an more efficient estimate of initial viral decay rates (i.e. estimate β_3 with smaller SE).
Better evaluate and compare different treatments since the initial viral decay rates may reflect the efficacies of the treatments. (DTP data analysis in Section 2)

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The advantages of modeling the variance

- Allow for inconstant variance, better understand the nature of the within-individual variation
- Obtain more efficient estimates (i.e., with smaller SE) of the main parameters of interest
- Robust inference to address outliers in the data

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Robust model

In the variance model (2), if we assume $\exp(a_i) \sim k/\chi^2_k$, it can be derived that

the within-individual error $e_{ij} \sim t(k)$,

where $t(k)$ is a t-distribution with degrees of freedom k.[2]

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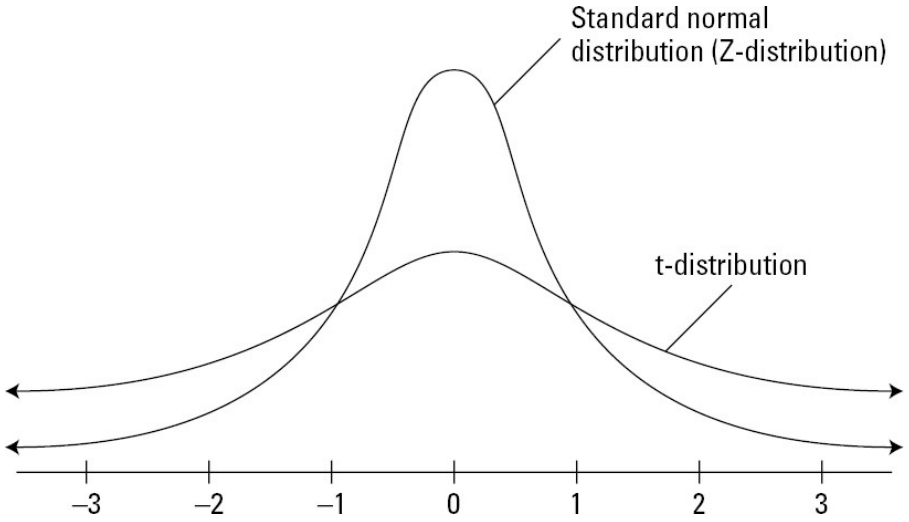

Figure: $t(k)$ has heavier tails, and thus it incorporates possible outliers

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Measurement error model

In the variance model (2), the covariate z_{ij} may be measured with errors. The covariate effect on the variance could be masked if the measurement errors are not addressed.

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Measurement error model

$$z_{ij} \equiv z_{ij}^* + \epsilon_{ij} = q(t_{ij}, \gamma, \boldsymbol{b}_i) + \epsilon_{ij} \tag{3}$$

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$$z_{ij} \equiv z_{ij}^* + \epsilon_{ij} = q(t_{ij}, \gamma, \boldsymbol{b}_i) + \epsilon_{ij} \tag{3}$$

- z_{ij}: observed covariate value; z_{ij}^*: unobserved true covariate value; ϵ_{ij}: measurement error

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- z_{ij}: observed covariate value; z_{ij}^*: unobserved true covariate value; ϵ_{ij}: measurement error
- $q(\cdot)$: a known linear function or a unknown smooth function

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- z_{ij}: observed covariate value; z_{ij}^*: unobserved true covariate value; ϵ_{ij}: measurement error
- $q(\cdot)$: a known linear function or a unknown smooth function

E.g., Model the CD4 process using a quadratic mixed effects model

$$z_{ij} = \gamma_0 + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2 + b_i + \epsilon_{ij}.$$

Or approximate the smooth function by using splines approach.

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- $q(\cdot)$: a known linear function or a unknown smooth function
- γ: fixed effects
- $\boldsymbol{b}_i$: random effects

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Real data analysis - DTP data

The Drug Treatment Program (DTP) in the British Columbia Centre for Excellence (BCCfE) provides the longitudinal viral load and CD4 cell count measurements during antiretroviral treatments (ART).

We are interested in

1. Studying the longitudinal virologic dynamics since ART initiation
2. Evaluating and comparing efficacy between different regimens by examining the initial viral decay rates

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Inclusion/Exclusion

The analytic data set includes

1. patients initiated treatment after April 01, 2011
2. follow-up time within the first year since the ART initiation before rebound
3. patients with 4 more more measurements in Both VL and CD4

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Inclusion/Exclusion

The analytic data set includes

1. patients initiated treatment after April 01, 2011
2. follow-up time within the first year since the ART initiation before rebound
3. patients with 4 more more measurements in Both VL and CD4

Result in 1796 patients with 11904 records in total.

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Sample description

- Median age of 40 at ART initiation; 87% males; 18% IDU
- 99% viral suppressed; 13% rebounded after suppression
- Median follow-up of 320 days (range from 73-365 days); re-scale time to $[0, 1]$
- The pharmacological classes within the first regimen

First regimen class	N	%
CAN/	5	0.28
IIN/	302	15.82
IIN/ENH/	113	6.29
NNR/	506	28.17
PI/ENH/	729	40.59
PI/IIN/ENH//	141	7.85

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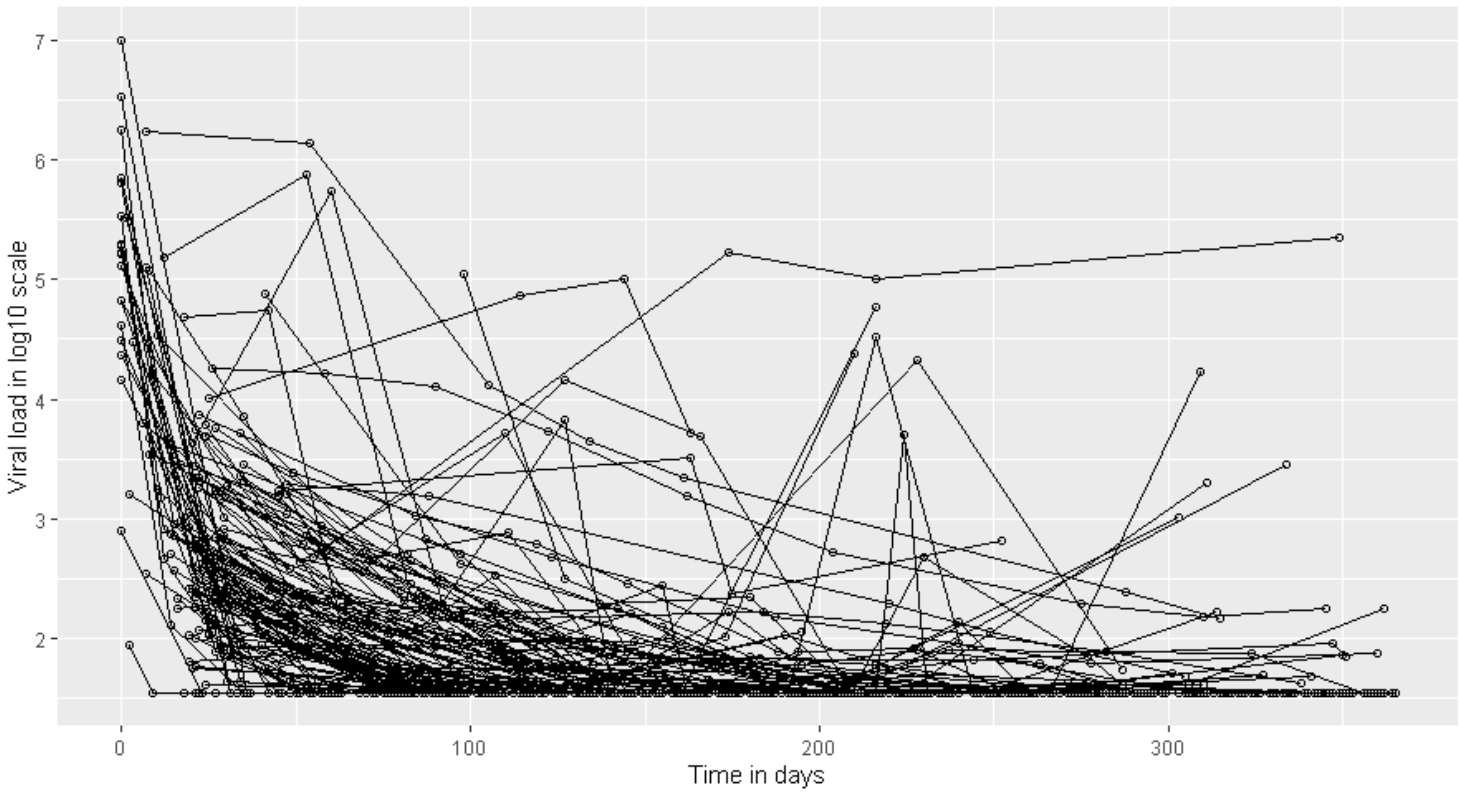
Viral load trajectories

Median number of VL measurements is 6 per patient, ranging from 4-15

Figure: Viral load (log10-) trajectories for 150 randomly selected patients



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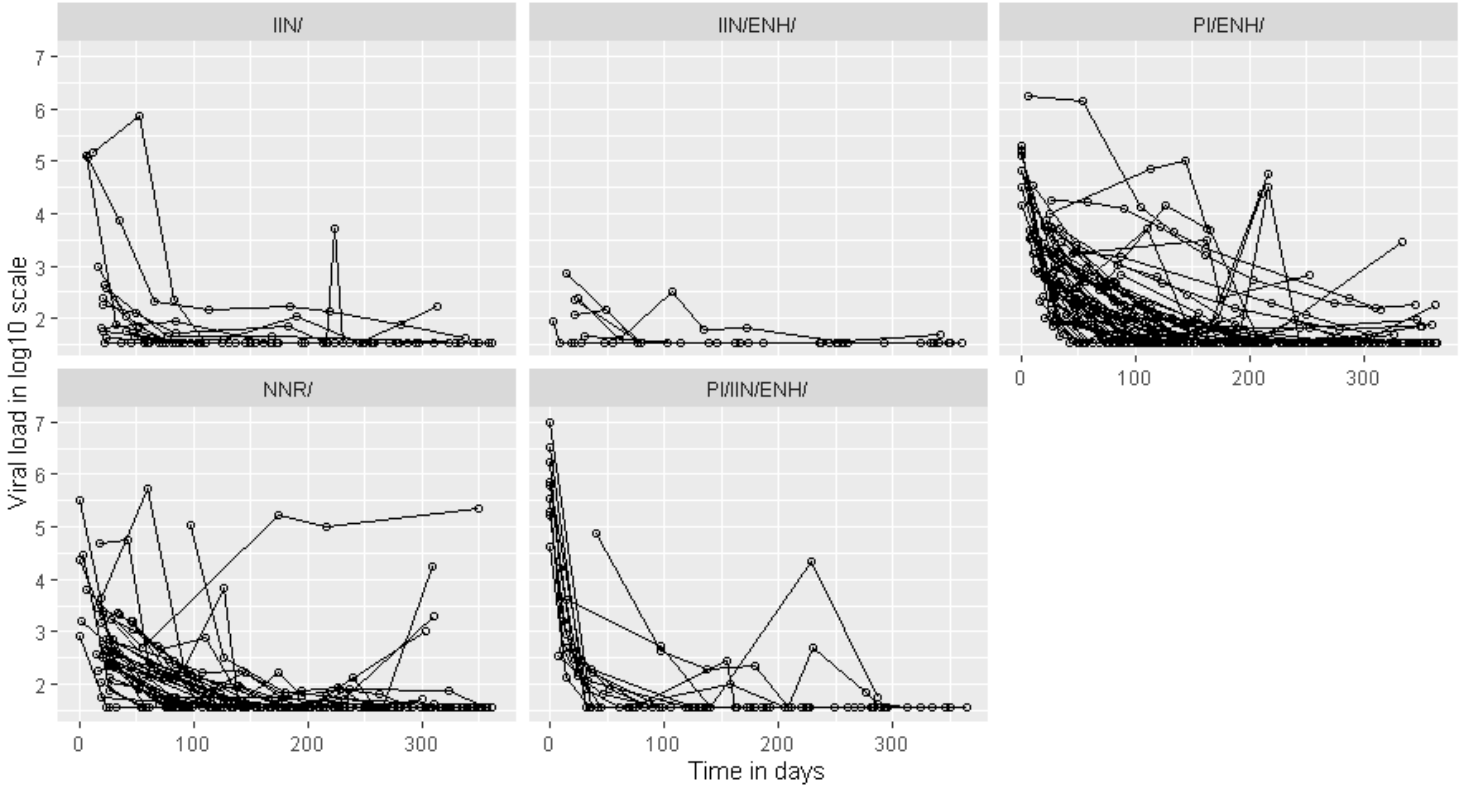
Viral load trajectories

Median number of VL measurements is 6 per patient, ranging from 4-15

Figure: Viral load (log10-) trajectories for 150 randomly selected patients, by first regimen class



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Use the proposed models to explore the initial viral decay rate

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Use the proposed models to explore the initial viral decay rate

- for each regimen class

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Use the proposed models to explore the initial viral decay rate

- for each regimen class
- modeling the within-individual variances with time-varying CD4 and random effect a_i

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Use the proposed models to explore the initial viral decay rate

- for each regimen class
- modeling the within-individual variances with time-varying CD4 and random effect a_i
- assuming $\exp(a_i) \sim k/\chi^2_k$ in the variance model to address possible outliers in the viral load measurements

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Use the proposed models to explore the initial viral decay rate

- for each regimen class
- modeling the within-individual variances with time-varying CD4 and random effect a_i
- assuming $\exp(a_i) \sim k/\chi^2_k$ in the variance model to address possible outliers in the viral load measurements
- considering measurement error model to address the measurement errors in the CD4 measurements

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NLME - the mean model

$$y_{ij} = (\beta_1 + u_{i1}) + (\beta_2 + u_{i2})e^{-(\beta_3 + u_{i3})t_{ij}} + e_{ij}$$

- y_{ij}: the $\log_{10}$ viral load measured at time t_{ij} for individual i
- β_3: initial viral decay rate
- $e_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$

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- y_{ij}: the $\log_{10}$ viral load measured at time t_{ij} for individual i
- β_3: initial viral decay rate
- $e_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$

NLME - the variance model

$$\log(\sigma_{ij}^2) = \alpha_1 + \alpha_2 z_{ij}^* + a_i$$

- z_{ij}^*: the "true" but unobservable (log-) CD4
- $\exp(a_i) \sim k/\chi_k^2$ for robust inference (pre-specify $k = 3$

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Measurement error model for CD4

$$z_{ij} \equiv z_{ij}^* + \epsilon_{ij} = (\gamma_0 + b_{i0}) + (\gamma_1 + b_{i1})t_{ij} + (\gamma_2 + b_{i2})t_{ij}^2 + \epsilon_{ij}$$

- z_{ij}: the observed (log-) CD4 with measurement error
- $(\gamma_0, \gamma_1, \gamma_2)$: fixed effects
- (b_{i0}, b_{i1}, b_{i2}): random effects

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Results: parameter estimates for two regimen classes

IIN/ENH regimen class:

Table: Model results for IIN/ENH regimen class

Model		IIN/ENH						
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1
Standard	Est	1.61	2.70	23.38				
	SE	0.027	0.154	2.166				
	p-val	$< .001$	$< .001$	$< .001$				
TS	Est	1.54	2.71	30.05	11.31	-2.41	6.08	0.22
	SE	0.005	0.056	1.451	0.699	0.113	0.062	0.043
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$
JM	Est	1.54	2.67	31.33	14.01	-2.89	6.06	0.28
	SE	0.003	0.047	1.590	0.744	0.121	0.061	0.032
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$

Standard: NLME without modeling variances. TS: two-step method: predict true CD4 first, and then fit NLME. JM: Joint modeling all models

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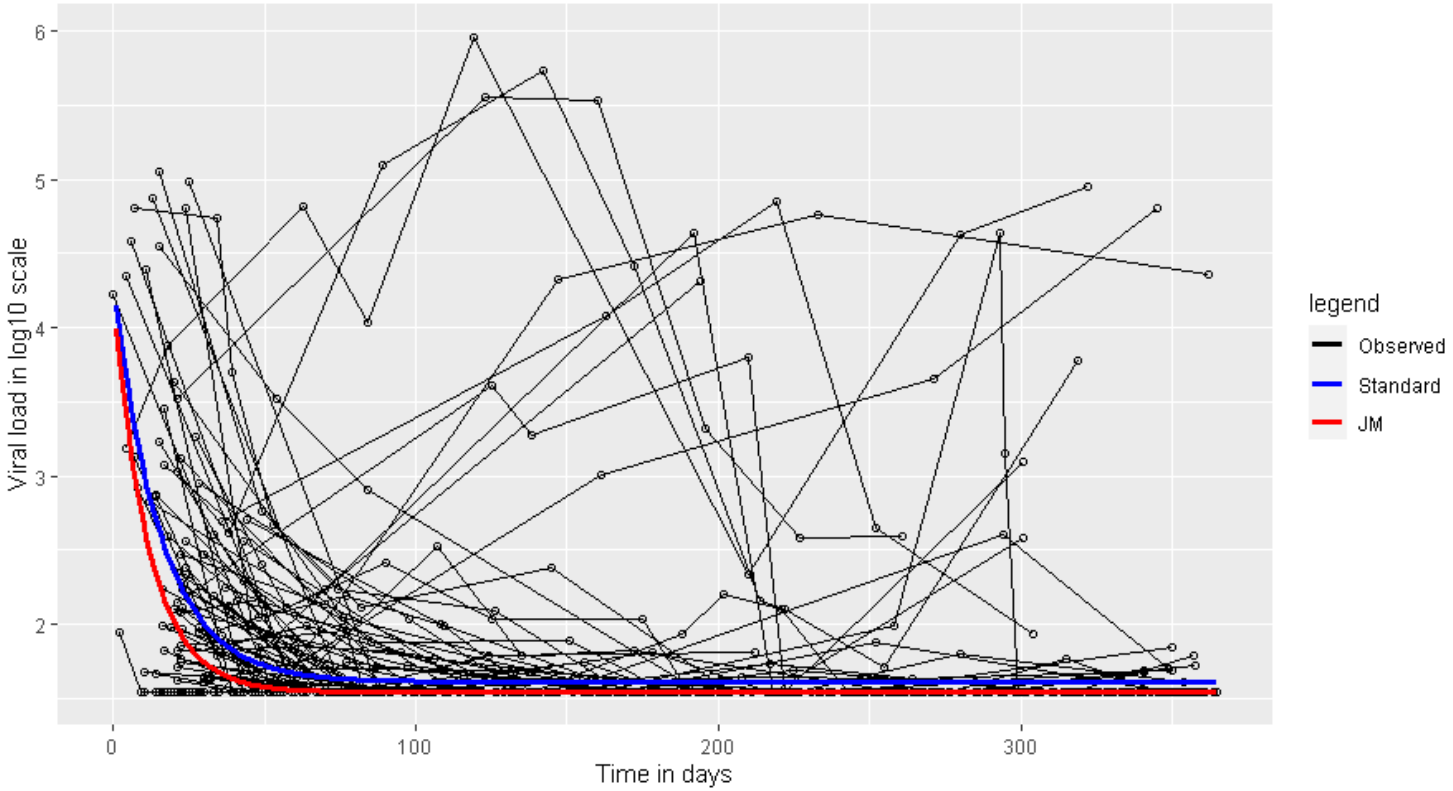
Results: parameter estimates for two regimen classes

IIN/ENH regimen class:

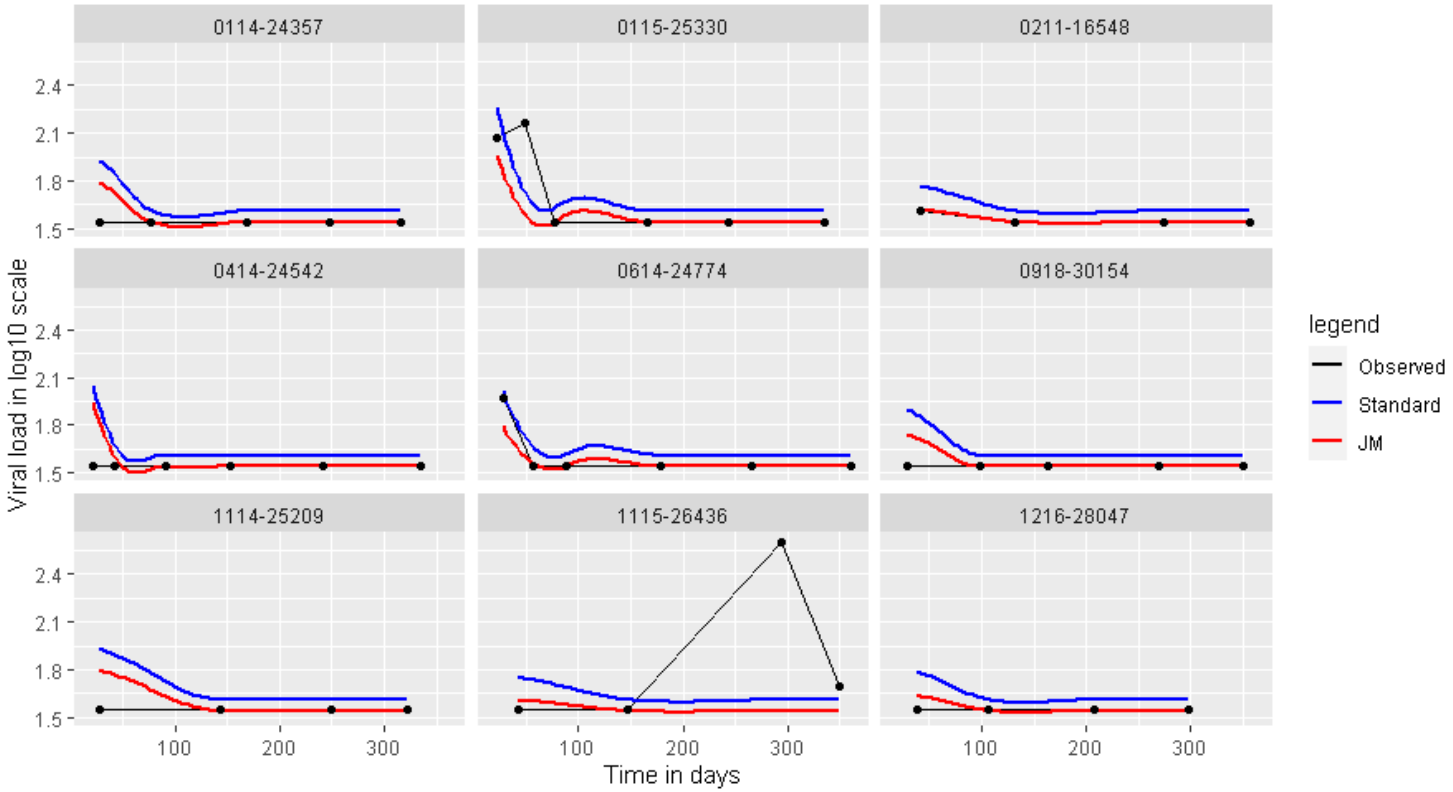
Figure: IIN/ENH regimen class: population average trajectory



Results: parameter estimates for two regimen classes

IIN/ENH regimen class:

Figure: IIN/ENH regimen class: Fitted trajectory for randomly selected patients



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Results: parameter estimates for two regimen classes

NNR/ regimen class:

Table: Model results for NNR/ regimen class

Model		NNR/							
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
Standard	Est	1.46	1.77	4.71					
	SE	0.013	0.063	0.263					
	p-val	$< .001$	$< .001$	$< .001$					
TS	Est	1.54	2.21	9.99	3.82	-1.16	5.89	0.69	-0.37
	SE	0.003	0.050	0.147	0.254	0.041	0.035	0.092	0.087
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	
JM	Est	1.54	2.22	10.03	4.44	-1.27	5.87	0.78	-0.43
	SE	0.003	0.050	0.149	0.250	0.041	0.032	0.056	0.053
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	

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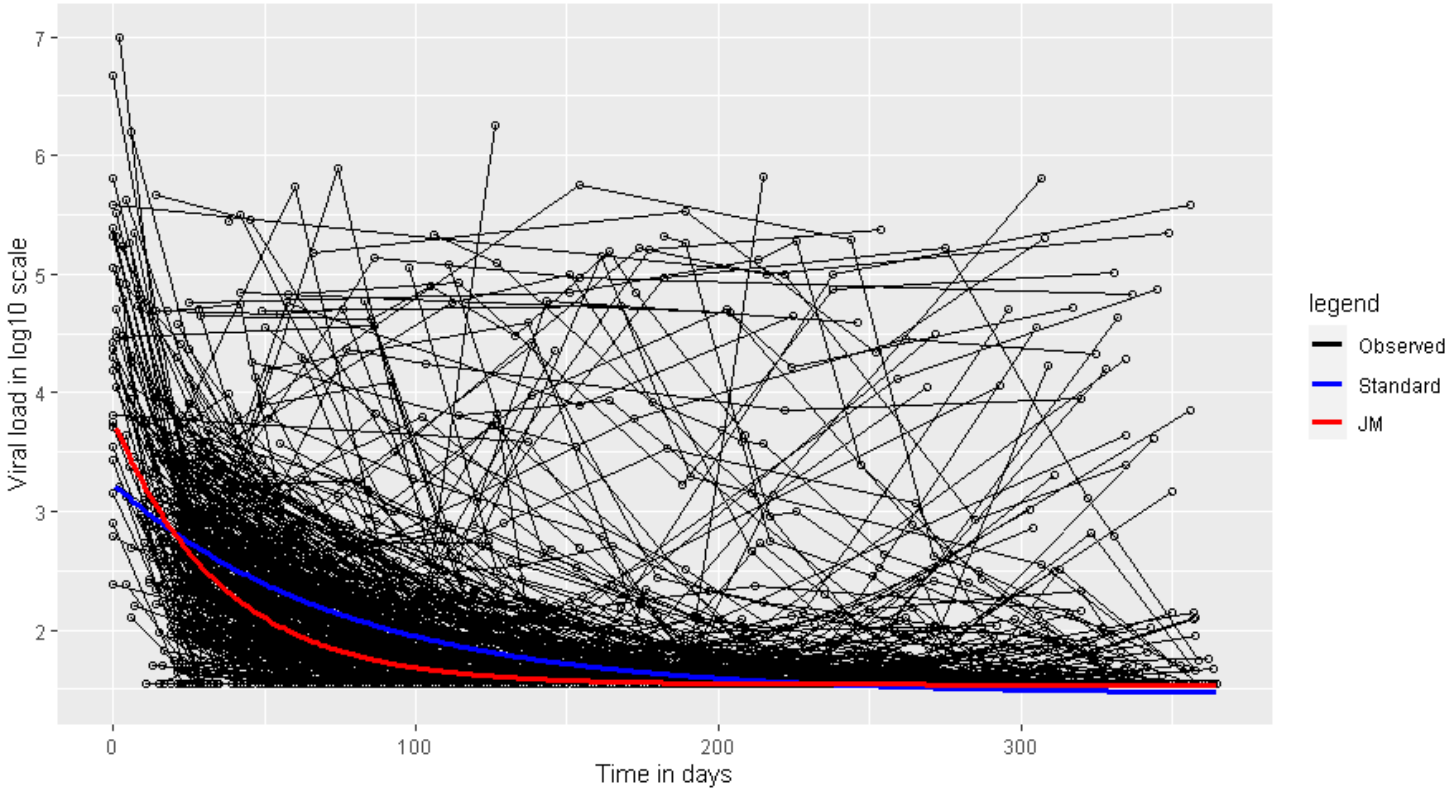
NNR/ regimen class:

Figure: NNR/ regimen class: population average trajectory



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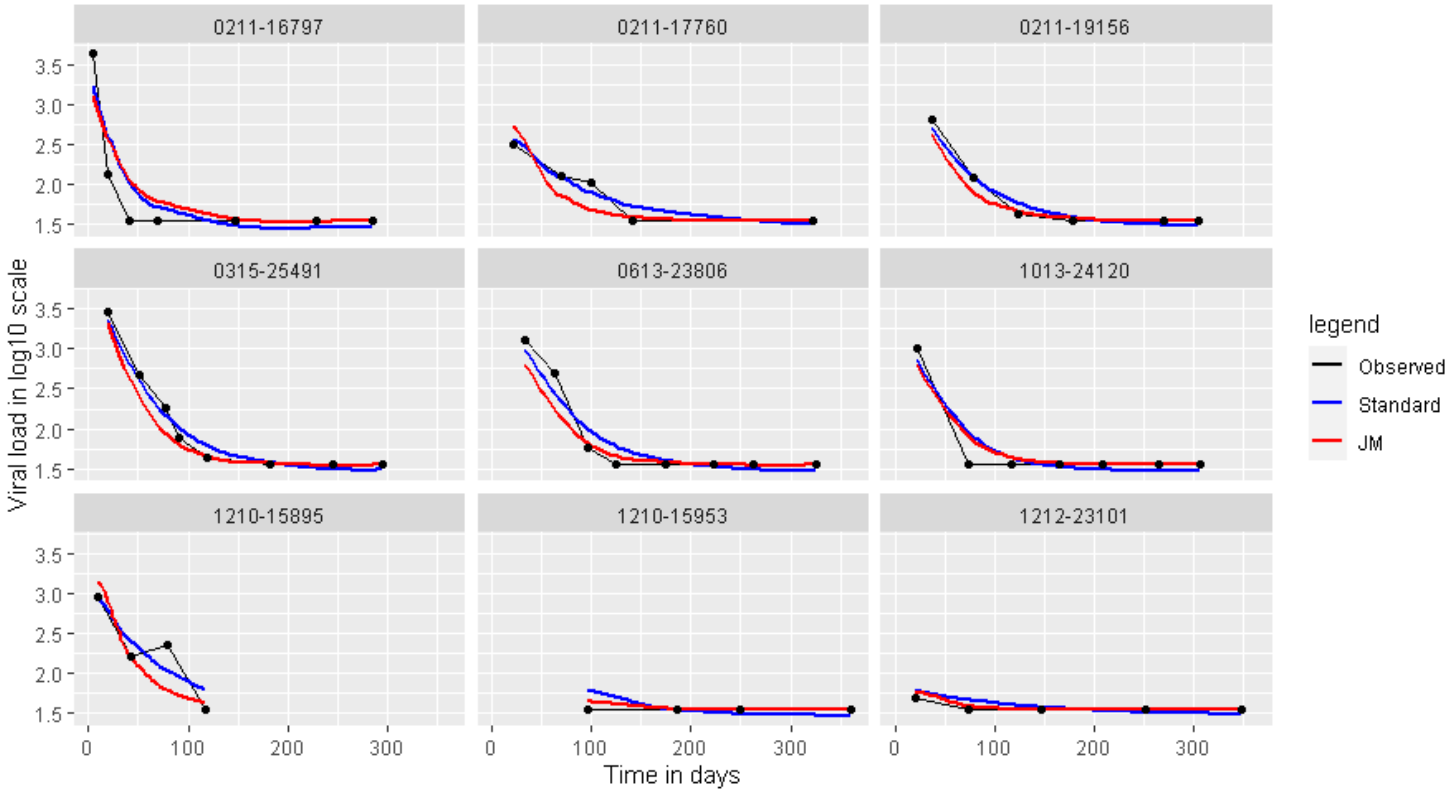

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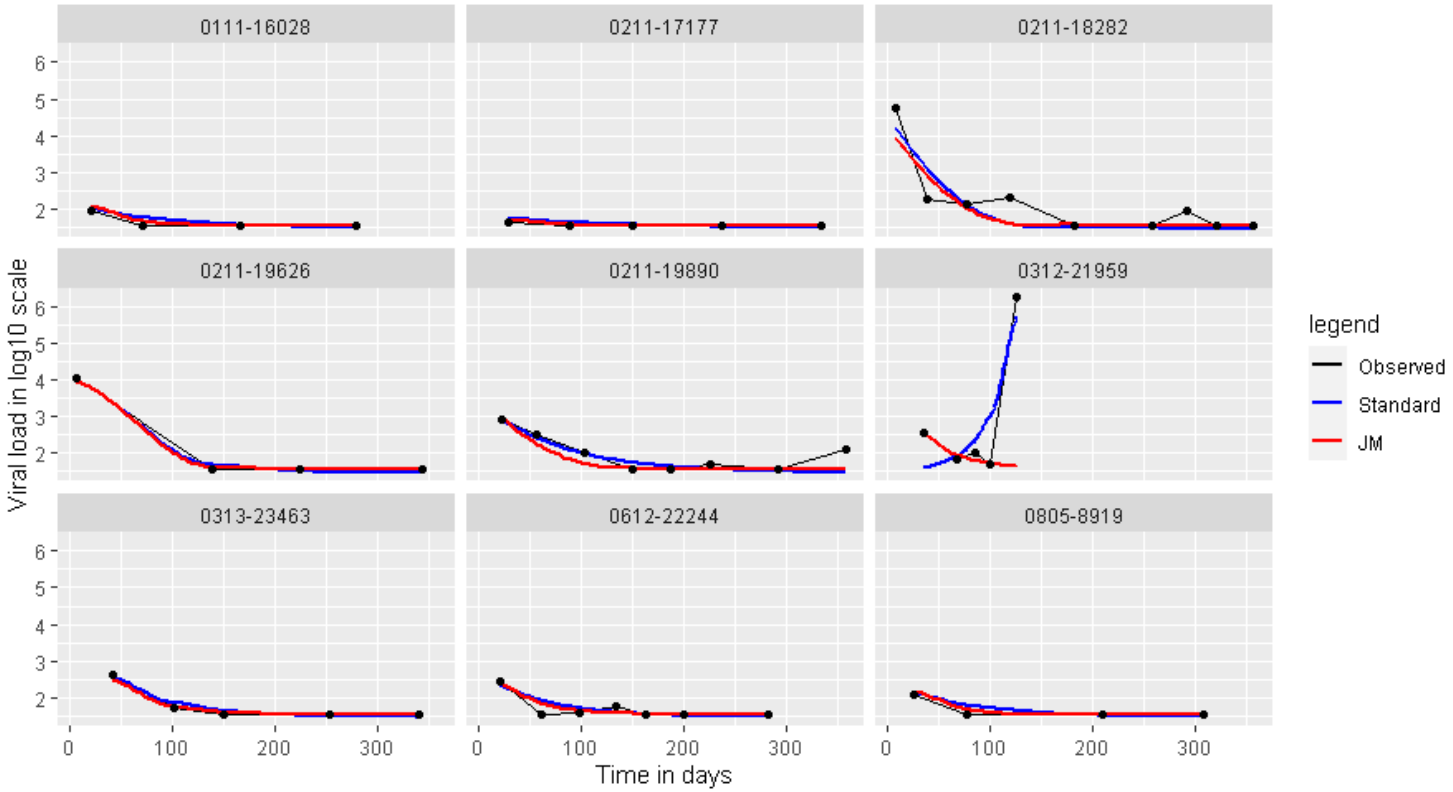
Results: parameter estimates for two regimen classes

NNR/ regimen class:

Figure: NNR/ regimen class: Fitted trajectory for randomly selected patients



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Simulation design

- n=100 individuals, with $n_i = 15$ repeated measurements for each individual.
- The entire study period is set to be [0, 1], and the 15 measuring time points are equally allocated
- The true parameter values are chosen based on real data analysis
- Consider two scenarios:
  - no outliers: generate within-individual errors using normal distribution
  - possible outliers: generate within-individual errors using $t(3)$ distribution
- The number of simulation repetition: 100

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Simulation design

Compare the parameter estimation between Standard, TS, and JM

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Simulation design

Compare the parameter estimation between Standard, TS, and JM , in term of

- "Estimate": average of the 100 estimates from the 100 simulation runs

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Simulation design

Compare the parameter estimation between Standard, TS, and JM , in term of

- "Estimate": average of the 100 estimates from the 100 simulation runs
- "rMSE": relative mean square error of the 100 parameter estimates

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  Defined as

$$\text{rMSE (\%) of } \hat{\beta} = \left| \frac{1}{100} \sum_{m=1}^{100} \left(\hat{\beta}^{(m)} - \beta \right)^2 / \beta \right| \times 100\%$$

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Simulation design

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- "Estimate": average of the 100 estimates from the 100 simulation runs
- "rMSE": relative mean square error of the 100 parameter estimates
- "SE": square root of the average of the 100 variance estimates

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- "Estimate": average of the 100 estimates from the 100 simulation runs
- "rMSE": relative mean square error of the 100 parameter estimates
- "SE": square root of the average of the 100 variance estimates
- "Coverage":empirical coverage based on SE among the 100 95% confidence intervals from the 100 simulation runs.

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- "Estimate": average of the 100 estimates from the 100 simulation runs
- "rMSE": relative mean square error of the 100 parameter estimates
- "SE": square root of the average of the 100 variance estimates
- "Coverage":empirical coverage based on SE among the 100 95% confidence intervals from the 100 simulation runs.
- "SE.bt": the average of bootstrapping SE from the 100 simulation runs (For JM estimation only)
- "Coverage.bt": empirical coverage based on bootstrapping SE (For JM estimation only)

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Scenario 1: No outlier

Table: Normal distribution for a_i: both data generation and modeling

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.527	2.996	8.054					
	SE	0.067	0.148	0.862					
	rMSE(%)	0.235	0.441	14.313					
	Coverage	0.920	0.990	0.910					
TS	Estimate	2.509	2.994	7.699	-8.392	1.547	5.206	1.598	-1.198
	SE	0.052	0.069	0.433	0.492	0.087	0.042	0.063	0.061
	rMSE(%)	0.102	0.222	3.647	8.426	1.252	0.033	0.207	0.258
	Coverage	0.960	0.920	0.930	0.760	0.800	0.980	0.970	0.960
JM	Estimate	2.511	2.995	7.759	-8.069	1.489	5.205	1.603	-1.202
	SE	0.052	0.069	0.441	0.486	0.086	0.042	0.060	0.058
	SE. bt	0.057	0.075	0.495	0.690	0.122	0.043	0.061	0.059
	rMSE(%)	0.106	0.222	4.027	7.479	1.267	0.033	0.193	0.237
	Coverage	0.950	0.930	0.920	0.800	0.810	0.980	0.970	0.960
	Coverage.bt	0.970	0.960	0.950	0.930	0.920	0.970	0.960	0.970

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Scenario 2: Possible outliers

Table: Inverse χ^2 distribution with 3 degree of freedom for $\exp(a_i)$: both data generation and modeling

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.490	3.007	7.546					
	SE	0.090	0.222	1.245					
	rMSE(%)	0.431	0.963	21.293					
	Coverage	0.920	0.990	0.950					
TS	Estimate	2.505	2.991	7.582	-8.024	1.552	5.199	1.605	-1.204
	SE	0.056	0.081	0.500	0.491	0.087	0.042	0.063	0.061
	rMSE(%)	0.162	0.242	5.636	6.973	1.368	0.043	0.254	0.328
	Coverage	0.920	0.940	0.850	0.810	0.820	0.940	0.960	0.960
JM	Estimate	2.507	2.991	7.645	-7.738	1.500	5.197	1.611	-1.208
	SE	0.055	0.081	0.507	0.486	0.086	0.042	0.060	0.058
	SE.bt	0.071	0.106	0.679	0.755	0.134	0.042	0.061	0.058
	rMSE(%)	0.163	0.241	6.023	8.892	1.363	0.043	0.245	0.316
	Coverage	0.910	0.940	0.850	0.740	0.780	0.940	0.920	0.910
	Coverage.bt	0.960	0.990	0.970	0.910	0.920	0.940	0.920	0.930

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